Common Dynamics in Non-Negative Panels with Many Zeros

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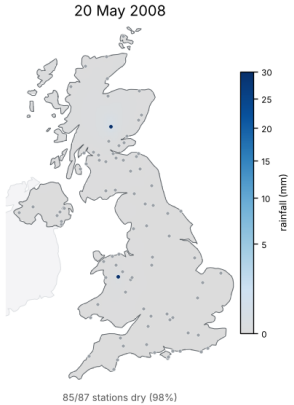
Outline

1. Motivation
2. The model
3. Assumptions
4. Theory
5. Estimation
6. Empirical illustration
7. Conclusion

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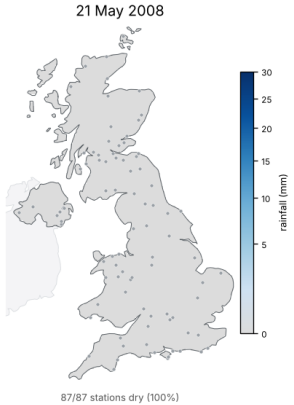
Step through days with → (20 May – 2 June 2008)



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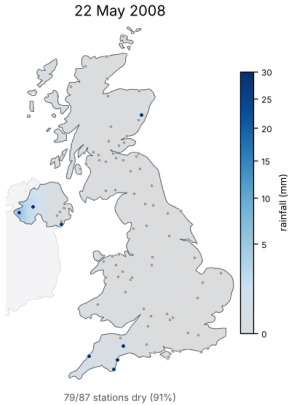
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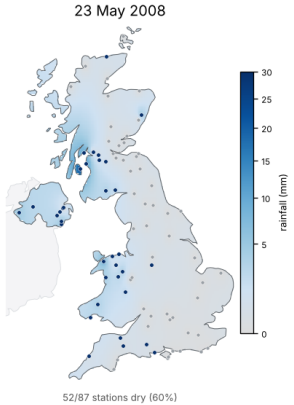
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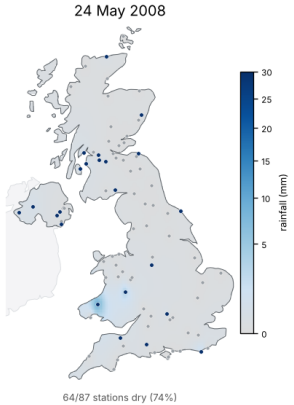
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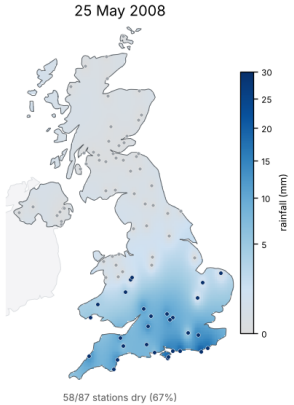
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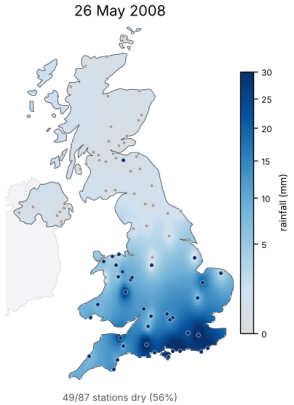
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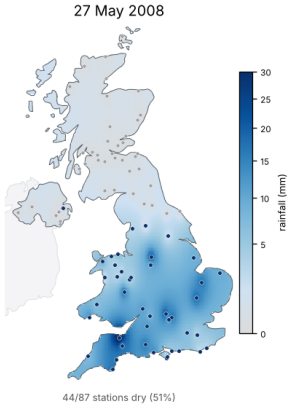
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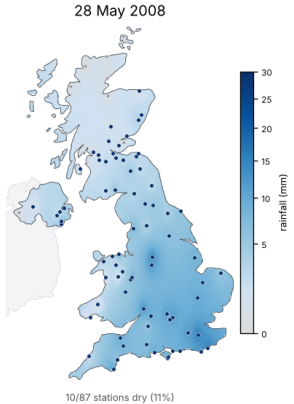
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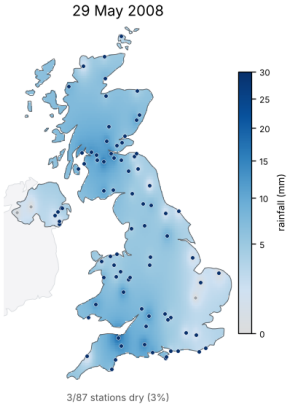
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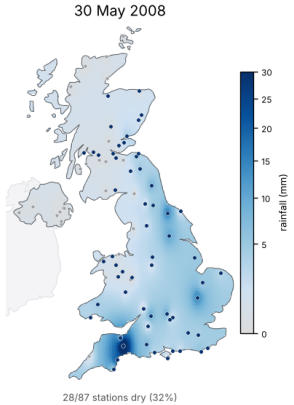
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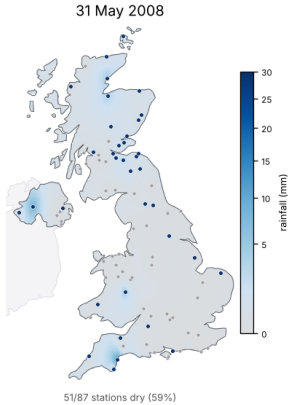
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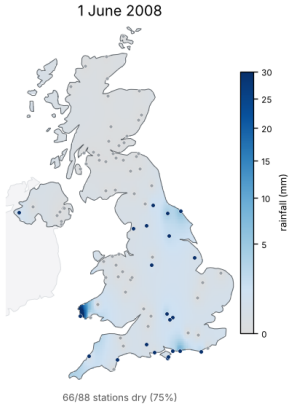
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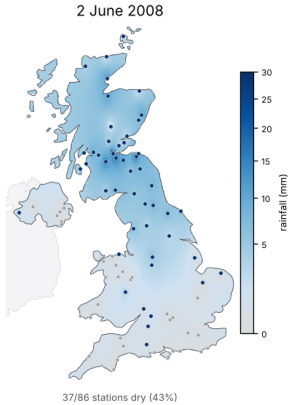
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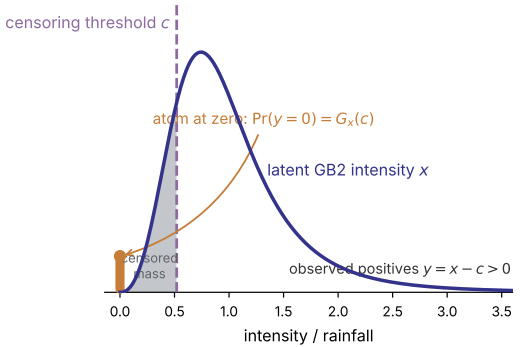
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A zero is informative, not missing

- What is recorded is a latent intensity that is always present.
- A zero marks that it fell below the level at which the measuring instrument, the market, or the reporting convention registers anything — a **detection / reporting threshold** c.
- A zero is therefore an **informative observation about the state** — to be modelled, not treated as missing.



Censored-score GB2 model (Harvey and Ito, 2020).

Comovement and joint modelling

- **Comovement** — these series move together: weather systems cross many stations at once, market-wide signals, ... so that zeros **cluster across the panel and over time**.

- **The questions of interest can be joint** — the probability a whole region is dry for a week, the tail of aggregate rainfall over a catchment, the risk that activity falls silent across a market, etc. — these concern the panel's **joint law**. (Gao et al., 2024)

Treating zeros

- **Drop the zeros** / condition on positives $\Rightarrow$ discards the informative feature (Santos Silva and Tenreyro, 2006; Cohn et al., 2022).
- **Add a constant**, $\log(y + c)$ $\Rightarrow$ estimates depend on an arbitrary c (Chen and Roth, 2024; Bellégo et al., 2022; Aïhounton and Henningsen, 2021; Mullahy and Norton, 2024).
- **Two-part / occurrence–amount** models split off the atom (Katz, 1977; Stern and Coe, 1984).
- **Multiplicative-error / ACD** give dynamics but place *no mass at zero* (Engle and Gallo, 2006; Engle and Russell, 1998).
- ***Censored score-driven GB2**: a latent GB2 censored at a threshold, filtered by the score, with a genuine **mass at zero** (Harvey and Ito, 2020).

A joint law for the panel: three routes

To model the panel jointly we need a **joint law**. Three possible routes:

Route	How the atom bites
Multivariate density	few constructions for censored densities, tied to a shared mixing variable, *not* observation-driven; the likelihood mixes a continuum with atoms on every coordinate subspace.
Copula (Sklar)	discontinuous margins $\Rightarrow$ copula *not unique*; margins & dependence no longer separate; biased rank/PIT estimators.
(Score-driven) dynamic factor	generates dependence directly, stays well-defined with atoms (✓) — but existing versions are elliptical only.

Yang et al. (2011); Sarabia et al. (2021); Sklar (1959); Genest and Nešlehová (2007); Creal et al. (2014); Artemova (2025); Harvey and Luati (2014).

The gap

	non-neg.	atom	heavy tail	asym.	panel
Harvey–Ito — univariate cGB2	✓	✓	✓	✓	—
Engle–Gallo — MEM / ACD	✓	—	✓	✓	—
Creal et al. — score-driven DFM	—	—	✓	✓	✓
Artemova — elliptical factor	—	—	✓	—	✓
This paper — joint-cGB2	✓	✓	✓	✓	✓

Our model — the idea in three steps

1. **Margins.** Each series is a **censored GB2** — non-negative, heavy-tailed, with a genuine **mass at zero**.
2. **Dependence.** All series' scales load on **the same common dynamic factors** — this is what makes it a **joint** law.
3. **Dynamics.** The factors are **score-driven** — each period, the score of the predictive likelihood updates them.

Measurement: the censored GB2

For series $i \in [N] = \{1, \ldots, N\}$ (fixed N) and time $t \in \{1, \ldots, T\}$ ($T \to \infty$) with $\mathcal{F}_{t-1} = \sigma(\mathbf{y}_1, \ldots, \mathbf{y}_{t-1})$, a latent $x_{it} \sim \mathrm{GB2}(e^{\phi_{i,t|t-1}}, \upsilon_i, \xi_i, \varsigma_i)$ is observed after censoring below at $c_i > 0$: $y_{it} = \max(x_{it} - c_i, 0)$.

- **Observation density** — continuous part on $y > 0$ and an atom at 0:

$$g_{it}(y) = \big[G_{x,it}(c_i)\big]^{\mathbb{1}(y=0)} \big[g_{x,it}(y + c_i)\big]^{\mathbb{1}(y>0)}, \qquad y \geq 0,$$

  where $g_{x,it}, G_{x,it}$ are the GB2 p.d.f. and c.d.f. at scale $e^{\phi_{i,t|t-1}}$.

- **Conditional probability of a zero**, driven by the log-scale (hence the factors):

$$\pi_{0,it} = \Pr(y_{it} = 0 \mid \mathcal{F}_{t-1}) = G_{x,it}(c_i),$$

  closed form and strictly decreasing in $\phi_{i,t|t-1}$ — more mass at zero as the scale $e^{\phi_{i,t|t-1}}$ falls.

GB2 properties we use (Lemma; proofs in Appendix)

Let $x \sim \mathrm{GB2}(e^{\phi}, \upsilon, \xi, \varsigma)$ and the beta transform $b(x) = \dfrac{(xe^{-\phi})^{\upsilon}}{1+(xe^{-\phi})^{\upsilon}} \in (0,1)$. Then:

(i) **Beta transform — the random shock.** $b(x) \sim \mathrm{Beta}(\xi, \varsigma)$: the $e^{-\phi}$ inside $b(\cdot)$ **cancels** x**'s scale**, so $b(x)$ is a *pivot* — its **law does not depend on** ϕ.
This b **is the random shock** $\mathbf{b}_t$ that drives the model: each outcome is a deterministic function of the state $\mathbf{f}$ and b_{it}, with b_{it} **free of the state**. So the recursion is an *iterated random function*, and $\mathbf{b}_t$ is the object we **hold fixed** across runs in the stationarity proof.

(ii) **Moments & tails.** $\mathbb{E}[x^r] < \infty \iff -\upsilon\xi < r < \eta$, with **tail index** $\eta := \upsilon\varsigma$; left-tail index $\upsilon\xi$ (pole at 0 iff $\upsilon\xi < 1$); every quantile scales by e^{ϕ}.

(iii) **Shape flexibility.** nests Singh–Maddala (ξ=1), Dagum (ς=1), log-logistic (ξ=ς=1), Lomax (υ=ξ=1); limits: generalized gamma, lognormal.

(iv) **Identification.** the censored GB2 family is globally identified (Proposition, App.).

The log-scale: common-factor structure

Each series' log-scale loads on $k \ll N$ common factors:

$$\phi_{i,t|t-1} = \omega_i + \mathbf{\Lambda}_i' \, \mathbf{f}_{t|t-1} + \mathbf{w}_t' \boldsymbol{\beta}_i \qquad \Longleftrightarrow \qquad \boldsymbol{\phi}_{t|t-1} = \boldsymbol{\omega} + \mathbf{\Lambda} \, \mathbf{f}_{t|t-1} + \boldsymbol{\beta} \, \mathbf{w}_t .$$

	dimension	role
$\boldsymbol{\omega}$	$N \times 1$	series intercepts, $\omega_i = \mathbb{E}[\phi_{i,t\|t-1}]$
$\mathbf{\Lambda} = [\mathbf{\Lambda}_1, \ldots, \mathbf{\Lambda}_N]'$	$N \times k$	factor loadings
$\mathbf{f}_{t\|t-1}$	$k \times 1$	common dynamic factors
$\boldsymbol{\beta}, \ \mathbf{w}_t$	$N \times P, \ P \times 1$	coefficients; deterministic seasonal basis

Parameter vector $\mathbf{\Theta} = \{\mathbf{c}, \boldsymbol{\upsilon}, \boldsymbol{\xi}, \boldsymbol{\varsigma}, \boldsymbol{\omega}, \boldsymbol{\beta}, \mathbf{\Lambda}, \mathbf{A}, \mathbf{B}\}$, of dimension $N(5 + P + k) + 2k$; in the application $\mathbf{w}_t$ is a cubic-spline seasonal basis with $P = 9$.

Factor transition — a score-driven AR

The factor is updated by the score-driven recursion

$$\mathbf{f}_{t+1|t} = \mathbf{A}\,\mathbf{f}_{t|t-1} + \mathbf{B}\,\tilde{\mathbf{u}}_t.$$

This is the shape of a vector AR(1) — a persistence term $\mathbf{A}\,\mathbf{f}$ plus an innovation $\mathbf{B}\,\tilde{\mathbf{u}}_t$ — except the innovation is **observation-driven**: $\tilde{\mathbf{u}}_t$ is the scaled score of today's likelihood, not an exogenous error.

$\mathbf{A}, \mathbf{B}$ are **diagonal**, so the recursion **decouples** into k scalar ones:

$$f_{j,t+1|t} = a_j\, f_{j,t|t-1} + b_j\, \tilde{u}_{j,t}, \qquad j = 1, \ldots, k.$$

Each factor is driven only by its own past and its own score — no cross-factor feedback in the dynamics (a_j: persistence of factor j; b_j: its responsiveness).

Score, information & scaled innovation (Lemma 1; proof in App.)

(1) Per-series score of the log-likelihood in the log-scale — a bounded martingale difference:

$$u_{it} = \frac{\partial \ln g_{it}(y_{it})}{\partial \phi_{i,t|t-1}} = \begin{cases} u_{it}^{+} = \upsilon_i(\xi_i + \varsigma_i)\, b_{it} - \upsilon_i \xi_i, & y_{it} > 0, \\ u_{it}^{0} = -\dfrac{\upsilon_i\, b_{c,it}^{\xi_i}(1 - b_{c,it})^{\varsigma_i}}{B(b_{c,it}; \xi_i, \varsigma_i)} < 0, & y_{it} = 0, \end{cases}$$

with $\mathbb{E}_{t-1}[u_{it}] = 0$ and $-\upsilon_i \xi_i < u_{it} < \upsilon_i \varsigma_i$.

(2) Factor space — stack $\mathbf{u}_t = (u_{1t}, \ldots, u_{Nt})'$, assuming conditional cross-sectional independence (A1), $\boldsymbol{\mathcal{I}}_t = \operatorname{diag}(\sigma^2_{u,1,t}, \ldots, \sigma^2_{u,N,t})$, $\sigma^2_{u,i,t} = \mathbb{E}_{t-1}[u^2_{it}]$:

$$\underbrace{\boldsymbol{\nabla}_t = \boldsymbol{\Lambda}' \mathbf{u}_t}_{\text{score}} \qquad \underbrace{\boldsymbol{\mathcal{I}}^f_t = \boldsymbol{\Lambda}' \boldsymbol{\mathcal{I}}_t \boldsymbol{\Lambda}}_{\text{Fisher information}} \qquad \underbrace{\tilde{\mathbf{u}}_t = (\boldsymbol{\mathcal{I}}^f_t)^{-\delta} \boldsymbol{\nabla}_t}_{\text{scaled innovation}}$$

with the score $\boldsymbol{\nabla}_t = \dfrac{\partial \ln g(\mathbf{y}_t \mid \boldsymbol{\phi}_{t|t-1})}{\partial \mathbf{f}_{t|t-1}}$ and conditional Fisher information $\boldsymbol{\mathcal{I}}^f_t = \mathbb{E}_{t-1}\big[\boldsymbol{\nabla}_t \boldsymbol{\nabla}_t'\big]$.

A censored score-driven dynamic factor model

The three components together — the model in one page:

- **Measurement.** $y_{it} = \max(x_{it} - c_i,\, 0), \qquad x_{it} \sim \mathrm{GB2}\big(e^{\phi_{i,t|t-1}}, \upsilon_i, \xi_i, \varsigma_i\big)$
- **Log-scale.** $\phi_{i,t|t-1} = \omega_i + \boldsymbol{\Lambda}_i' \, \mathbf{f}_{t|t-1} + \mathbf{w}_t' \boldsymbol{\beta}_i$
- **Factor transition.** $\mathbf{f}_{t+1|t} = \mathbf{A} \, \mathbf{f}_{t|t-1} + \mathbf{B} \, \tilde{\mathbf{u}}_t;\ \tilde{\mathbf{u}}_t = \big(\boldsymbol{\mathcal{I}}_t^f\big)^{-\delta} \boldsymbol{\nabla}_t.$

Assumptions I — model and regularity

A1 **(Cross-sectional independence).** Given $\mathcal{F}_{t-1}$, $x_{1t}, \ldots, x_{Nt}$ are independent, $x_{it} \sim \mathrm{GB2}(e^{\phi_{i,t|t-1}}, \upsilon_i, \xi_i, \varsigma_i)$; $\mathbf{w}_t$ is $\mathcal{F}_{t-1}$-measurable; $c_i > 0$. The panel log-likelihood is then **exact**:

$$\mathcal{L}_T(\boldsymbol{\Theta}) = \sum_{t=1}^{T} \sum_{i=1}^{N} \ln g_{it}\big(y_{it} \mid \mathcal{F}_{t-1}; \boldsymbol{\Theta}\big).$$

A2 **(Parameter space).** $\boldsymbol{\Theta}_0$ is interior to a compact $\mathcal{P}$; shapes $(\upsilon_i, \xi_i, \varsigma_i)$ and thresholds c_i finite and > 0; $(\omega_i, \boldsymbol{\beta}_i, \boldsymbol{\Lambda}_i)$ bounded; $\mathbf{A}, \mathbf{B}$ diagonal with $\max_j |a_j| \le \bar{a}$, $\max_j |b_j| \le \bar{b}$.

A3 **(Covariates).** $\{\mathbf{w}_t\}$ deterministic and bounded, with $T^{-1} \sum_t \mathbf{w}_t \to \mathbf{0}$.

A4 **(Information floor).** if $\delta > 0$: $\boldsymbol{\mathcal{I}}_t^f(\boldsymbol{\Theta}) \succeq \iota \, \mathbf{I}_k$ for some $\iota > 0$, all t and $\boldsymbol{\Theta} \in \mathcal{P}$. ▸ why A4?

Assumptions II — stability and identification

A5 **(Contraction).** (a) filter: $\bar{a} + \bar{b}\, L_\delta < 1$; (b) *mean:* $\bar{a} + \bar{b}\, M_\delta < 1$ — with L_δ, M_δ the Lipschitz and mean-Lipschitz moduli of the scaled innovation $\tilde{\mathbf{u}}_t$.

A6 **(Identification).** Let $\boldsymbol{\Omega} = \lim_T T^{-1} \sum_t \mathbb{E}[(1, \mathbf{w}_t', \mathbf{f}_{t|t-1}')'(1, \mathbf{w}_t', \mathbf{f}_{t|t-1}')]$; ℓ_t the time-t log-likelihood. Then, (a) $\operatorname{rank}(\boldsymbol{\Lambda}) = k$ and $\boldsymbol{\Omega}$ is non-singular; (b) distinct dynamics $(a_j, b_j) \neq (a_\ell, b_\ell)$ for $j \neq \ell$; with normalizations (N1)–(N3). ▸ identification ▸ N1–N3

A7 **(Information).** $\boldsymbol{\Theta}_0$ is interior and $\mathcal{H}(\boldsymbol{\Theta}_0) = -\mathbb{E}[\partial^2 \ell_t(\boldsymbol{\Theta}_0)/\partial\boldsymbol{\Theta}\, \partial\boldsymbol{\Theta}']$ is non-singular.

What we prove

Under A1–A7, the MLE $\widehat{\boldsymbol{\Theta}}_T$ satisfies four results:

1. **Invertibility** — the filter forgets its initialization (data held fixed). *standard*
2. **Stationarity & ergodicity** of $\{\mathbf{f}_t\}$ — the atom makes the data-generating map **discontinuous**. *the hard step*
3. **Consistency** — $\widehat{\boldsymbol{\Theta}}_T \to \boldsymbol{\Theta}_0$ almost surely. *standard*
4. **Asymptotic normality** — $\sqrt{T}(\widehat{\boldsymbol{\Theta}}_T - \boldsymbol{\Theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{H}(\boldsymbol{\Theta}_0)^{-1})$. *standard*

All four turn on one **common tool — a contraction**. For (1),(3),(4) it is the textbook *pathwise* contraction; for (2) the atom breaks that and we use a **contraction in mean**. We take each in turn, in detail on (2).

A compact factor space — the stage for contraction

The factor follows the score-driven recursion $\mathbf{f}_{t+1|t} = \mathbf{A}\,\mathbf{f}_{t|t-1} + \mathbf{B}\,\tilde{\mathbf{u}}_t$. The **scaled innovation** is bounded:

$$\|\tilde{\mathbf{u}}_t\| \;\le\; \iota^{-\delta}\,\|\mathbf{\Lambda}\|_2\sqrt{N}\,\bar{U} \;=:\; \bar{u} \;<\; \infty \quad \text{a.s.}$$ ▸ bounded score

Then, we prove that

Proposition (Compact factor space)

Under A1–A4 with $\bar{a} < 1$, the factor recursion admits an absorbing ball

$$\mathcal{X} = \{\,\mathbf{f} :\; \|\mathbf{f}\| \le R\,\}, \qquad R = \frac{\bar{b}\,\bar{u}}{1-\bar{a}},$$

with $\mathbf{f}_{t|t-1} \in \mathcal{X}$ for all t whenever $\mathbf{f}_{1|0} \in \mathcal{X}$. From any start,

$$\|\mathbf{f}_{t|t-1}\| \le \bar{a}^{\,t-1}\|\mathbf{f}_{1|0}\| + R\,(1-\bar{a}^{\,t-1}) \to R$$

.

Contraction

Write the factor update as $\mathbf{f}_{t+1} = \Phi_t(\mathbf{f}_t)$, a (random) map on the compact set $\mathcal{X}$.

Definition (Contraction)

Φ_t is a **contraction with modulus** $\rho < 1$ if it brings any two states closer:

$$\|\Phi_t(\mathbf{f}) - \Phi_t(\mathbf{f}')\| \;\leq\; \rho\,\|\mathbf{f} - \mathbf{f}'\|, \qquad \rho < 1$$

$\mathbf{f}, \mathbf{f}'$: two candidate states; $\|\cdot\|$: Euclidean distance; ρ: the contraction modulus.

Consequence. iterating, $\|\mathbf{f}_t - \mathbf{f}'_t\| \leq \rho^t \|\mathbf{f}_0 - \mathbf{f}'_0\| \to 0$: the start is forgotten geometrically and a unique stable path exists (Banach).

For our model, $\Phi_t : \mathbf{f} \mapsto \mathbf{A}\mathbf{f} + \mathbf{B}\,\tilde{\mathbf{u}}_t(\mathbf{f})$, so $\rho = \bar{a} + \bar{b}\,L$, where $\bar{a} = \max_j |a_j|$ (persistence, $\mathbf{A}$), $\bar{b} = \max_j |b_j|$ (responsiveness, $\mathbf{B}$), and L measures how strongly $\tilde{\mathbf{u}}_t$ responds to $\mathbf{f}$.

Two directions, one contraction

The same recursion can forget its initialization in two different directions:

	Invertibility (easy)	**Stationarity** (hard)
Question	recover the factor from data?	does the model have a stable law?
Hold fixed	the data $\mathbf{y}$	the shocks $\mathbf{b}$ (beta transforms)
Vary	the initial guess	the initial state
Two runs	same data, different seeds	same shocks, different starts
Map in $\mathbf{f}$	**smooth** (wet/dry frozen)	**jumps** (state crosses threshold)
Contraction	pathwise, $\rho_L = \bar{a} + \bar{b}\, L_\delta$	in mean, $\rho = \bar{a} + \bar{b}\, M_\delta$

Like an ARMA(1,1): **invertibility** is the *MA-root* condition (back out the innovations); **stationarity** is the *AR-root* condition (a stable law) — two separate properties.

The easy direction — invertibility

Setup. Hold the data fixed and run the filter from two seeds $\widehat{\mathbf{f}}_1 \neq \mathbf{f}_1$. Φ_t is the filter update.

Why it contracts. With the data fixed the indicator $\mathbb{1}(y_{it} > 0)$ is constant, so Φ_t is smooth and its response is a bounded pathwise slope:

$$L_\delta = \sup_{\mathbf{f}} \left\| \partial \tilde{\mathbf{u}}_t / \partial \mathbf{f}' \right\| < \infty \qquad \Longrightarrow \qquad \rho_L = \bar{a} + \bar{b}\, L_\delta .$$

Plug into the definition (with $\rho_L < 1$, Assumption A5(a)):

$$\boxed{\| \widehat{\mathbf{f}}_{t+1|t} - \mathbf{f}_{t+1|t} \| \;\leq\; \rho_L \, \| \widehat{\mathbf{f}}_{t|t-1} - \mathbf{f}_{t|t-1} \| \;\leq\; 2R\, \rho_L^{\,t-1} \;\longrightarrow\; 0}$$

The filter is **invertible**. L_δ is a pathwise slope, so this is the smooth case standard score-driven theory already covers (Straumann and Mikosch, 2006; Blasques et al., 2022).

The hard direction — why the pathwise route fails

Now vary the state, hold the shocks fixed: the map is the data-generating recursion $\mathbf{f} \mapsto F_t(\mathbf{f}, \mathbf{b})$, and the factor now enters the indicator. A pathwise contraction would need, for every shock $\mathbf{b}$,

$$\|F_t(\mathbf{f}, \mathbf{b}) - F_t(\mathbf{f}', \mathbf{b})\| \;\leq\; \rho \, \|\mathbf{f} - \mathbf{f}'\|$$

But the map jumps. $F_t(\mathbf{f}, \mathbf{b}_t) = \mathbf{A}\mathbf{f} + \mathbf{B}\,\tilde{\mathbf{u}}_t(\mathbf{f}, \mathbf{b}_t)$. As $\mathbf{f}' \to \mathbf{f}$ *across* a series' threshold, that day flips across wet and dry, and the score jumps across u^+ and u^0, so the left side stays bounded away from 0 while the right side $\to 0$:

$$\|F_t(\mathbf{f}, \mathbf{b}) - F_t(\mathbf{f}', \mathbf{b})\| \;\not\leq\; \rho \, \|\mathbf{f} - \mathbf{f}'\| \qquad \text{(no finite } \rho\text{)}.$$

The map is **discontinuous** in $\mathbf{f}$: no pathwise modulus exists. Classical SRE / Bougerol theory (Bougerol, 1993; Straumann and Mikosch, 2006) needs a continuous map.

The fix — a contraction in mean

Same definition, averaged over the shock. We cannot bound the jump for every $\mathbf{b}$, but we can bound it **on average**:

$$\boxed{\mathbb{E}_{\mathbf{b}}\big\|F_t(\mathbf{f},\mathbf{b}) - F_t(\mathbf{f}',\mathbf{b})\big\| \;\le\; \rho\,\|\mathbf{f}-\mathbf{f}'\|, \qquad \rho = \bar{a} + \bar{b}\,M_\delta < 1}$$

with M_δ the **in-mean** response of $\tilde{\mathbf{u}}_t$ to $\mathbf{f}$ (the averaged analogue of L_δ).

- The beta shocks $\mathbf{b}_t$ are i.i.d. and state-independent (GB2 property), so the recursion is an **iterated random function**. For these, an average contraction is a standard route to ergodicity (Diaconis and Freedman, 1999; Hairer et al., 2011; Wu and Shao, 2004; Meyn and Tweedie, 2009).
- It gives a **Wasserstein-1 contraction**: a unique invariant law — strictly stationary, ergodic, all moments finite.

Why the mean contracts — reduce to the score

The scaled innovation is $\tilde{\mathbf{u}}_t = (\mathcal{I}_t^f)^{-\delta}\mathbf{\Lambda}'\mathbf{u}_t$. Its mean sensitivity to the state splits in two:

- the **information** $(\mathcal{I}_t^f)^{-\delta}$ is a *smooth*, Lipschitz function of $\mathbf{f}$ — the floor (A4) keeps $x \mapsto x^{-\delta}$ well-behaved;
- the only non-smooth piece is the **per-series score** $u_i(\mathbf{f})$, which **jumps** where a day switches wet $\leftrightarrow$ dry.

So the whole contraction reduces to **one estimate** — a *mean*-Lipschitz bound on the score:

$$\boxed{\mathbb{E}_{\mathbf{b}}\left|u_i(\mathbf{f}) - u_i(\mathbf{f}')\right| \;\le\; C_i\,|\phi_i - \phi_i'|}, \qquad \phi_i = \omega_i + \mathbf{\Lambda}_i'\mathbf{f} + \mathbf{w}_t'\boldsymbol{\beta}_i.$$

We get it by **coupling**: feed the *same* shock $\mathbf{b}_t$ to both states. The two states can disagree only in a thin window.

▸ details

The key estimate — a jump, rare in proportion

Under that coupling, compare the two states on series i — two cases:

- **same branch** (both wet / both dry): u_i is smooth, so $|u_i(\mathbf{f}) - u_i(\mathbf{f}')| \le K_i^0 \, |\Delta\phi_i|$ for *every* shock.
- **flip** (one wet, one dry): u_i **jumps**, by at most $2\bar{U}_i$. But a flip needs the shock to land *between* the two thresholds, and that probability is the **change in the dry-day probability**

$$\Pr(\text{flip}) = \left|\pi_{0,i}(\mathbf{f}) - \pi_{0,i}(\mathbf{f}')\right| \;\le\; \bar{U}_i \, |\Delta\phi_i|,$$

  small because $\pi_{0,i} = G_{x,i}(c_i)$ is *smooth* in ϕ_i.

Pathwise the jump is discontinuous; **averaged** it is (jump)×(prob)$= 2\bar{U}_i^2|\Delta\phi_i|$. Hence

$$\mathbb{E}_{\mathbf{b}} \, |u_i(\mathbf{f}) - u_i(\mathbf{f}')| \;\le\; \Big(\underbrace{K_i^0}_{\text{smooth}} + \underbrace{2\bar{U}_i^2}_{\text{jump} \times \text{prob}} \Big) |\Delta\phi_i| \implies M_\delta < \infty, \quad \rho = \bar{a} + \bar{b}\, M_\delta < 1 \text{ (A5b)}.$$

Consistency & asymptotic normality

- **Consistency.** stationarity + invertibility + identification + a mild envelope $(\mathbb{E}|\ln x_{it}| < \infty) \Rightarrow \ \widehat{\boldsymbol{\Theta}}_T \to \boldsymbol{\Theta}_0$ a.s.
- **Asymptotic normality.** with $\delta > 0$, $\boldsymbol{\Theta}_0$ interior and $\mathcal{H}(\boldsymbol{\Theta}_0)$ non-singular,

$$\sqrt{T}(\widehat{\boldsymbol{\Theta}}_T - \boldsymbol{\Theta}_0) \ \xrightarrow{d} \ N\big(\mathbf{0},\ \mathcal{H}(\boldsymbol{\Theta}_0)^{-1}\big),$$

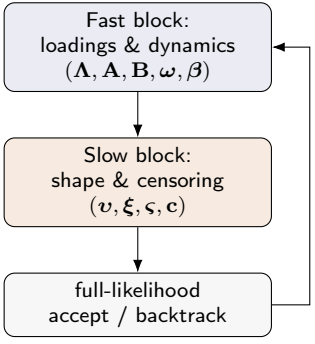
  standard once (1),(2) hold (Straumann and Mikosch, 2006; Blasques et al., 2022).
- **Cross-sectional dependence.** dropping A1, $\mathcal{L}_T$ is read as a composite likelihood; the estimator stays consistent (**QMLE**) with a **sandwich** covariance $\mathcal{H}^{-1}\mathcal{V}\mathcal{H}^{-1}$ — factors and forecasts unchanged, only standard errors widen. ▸ details

Estimation — a three-stage pipeline

Closed-form panel log-likelihood; MLE $\widehat{\boldsymbol{\Theta}}_T = \arg\max_{\boldsymbol{\Theta}\in\mathcal{P}} \mathcal{L}_T$. Direct joint maximization is high-dimensional, so we proceed in stages.

1. **Univariate warm-up.** Fit N per-series censored-GB2 models (Harvey and Ito, 2020) $\Rightarrow$ starts for shapes, thresholds, intercepts, seasonals.
2. **Number of factors k.** Eigenvalue criteria (Onatski, 2010; Ahn and Horenstein, 2013) on the de-seasonalized univariate log-scales ($k = 1$ here).
3. **Joint blockwise MLE.** Alternate a **fast block** $(\boldsymbol{\Lambda}, \mathbf{A}, \mathbf{B}, \boldsymbol{\omega}, \boldsymbol{\beta})$ and a **slow block** (shapes, c_i); monotone acceptance in the full recursive likelihood.

Stage 3: block coordinate ascent



In the spirit of feasible large-factor estimation (Doz et al., 2011).

Estimation — Blockwise MLE

Start: admissible parameters from Stages 1–2; keep $\mathbf{f}_{1|0} = \mathbf{0}$ fixed throughout. Run the filter.

1. **Fast block: hold shapes and censoring $\mathbf{c}$ fixed; repeat:**
   - **Location/loadings:** hold the current factor path and $(\mathbf{A}, \mathbf{B})$ fixed. Propose $(\boldsymbol{\omega}, \boldsymbol{\beta}, \boldsymbol{\Lambda})$ via MLE; normalize loadings under (N1)–(N3).
   - **Dynamics:** hold the proposed $(\boldsymbol{\omega}, \boldsymbol{\beta}, \boldsymbol{\Lambda})$ fixed. Update $(\mathbf{A}, \mathbf{B})$ via MLE (recomputing the entire factor path and likelihood for each candidate value).
   - Refilter and accept/backtrack using the full likelihood; refresh the factor path and repeat until inner convergence.
2. **Slow block: hold all fast-block parameters fixed.** Also freeze their current log-scale path; propose $(\upsilon_i, \xi_i, \varsigma_i)$ and c_i series by series via MLE. Refilter and accept/backtrack using the full likelihood.
3. **Cycle and stop.** With the accepted shapes and $\mathbf{c}$ fixed, reconverge the fast block. Stop when likelihood and parameters stabilize (or at the iteration limit).

Paths are frozen only for proposals, never for acceptance

Refilter; require admissibility, a valid filter and $\mathcal{L}_T(\text{new}) \geq \mathcal{L}_T(\text{current})$. Otherwise backtrack or keep the current value.

Data: UK daily rainfall — summary statistics

MIDAS Open (Met Office, 2023): $N = 88$ stations, 2003–2022, $T = 7{,}305$ days ($642{,}840$ cells, LOCF-balanced). 39.1% of station-days are dry.

	Min	Q_{25}	Median	Mean	S.d.	Q_{75}	Q_{90}	Q_{99}	Max
Dry-day proportion (%)	19.5	32.9	38.7	39.1	8.9	45.6	50.8	55.3	56.7
Wet-day rainfall (mm)	0.10	0.70	2.70	10.50	29.83	7.70	18.20	159.2	377.3

Zero-proportion group	Low (wet)	Moderate	High (dry)
Number of stations	22	42	24
Median dry-day prop.	0.284	0.384	0.504

Setup — three models, one panel

What we compare — and the role each one plays:

- **joint-cGB2** *(ours)* — one common score-driven factor, censored-GB2 margins: a genuine **joint law** for the panel.
- **indv-cGB2** — the same censored-GB2 margin fitted **station by station**, no cross-sectional link. The **stacked individual**: it fits each series as well as a univariate cGB2 can ($\approx 88\times$ the log-scale parameters) but delivers **no joint law**.
- Student-t — the one-factor **elliptical** benchmark (Artemova, 2025): same factor structure, a Student-t in place of the censored GB2. The only competing *joint* model.

Metrics. NLL: density (panel-pooled and typical-station); **PIT variance**: calibration (ideal $1/12$); **zero-Brier skill**: dry-day probability versus climatology.

Setup — scheme

1. In-sample scheme. Estimate parameters and evaluate the fitted model using the full sample.

2. Out-of-sample scheme.

1. **Training and refitting.** Use a rolling **5-year window**; refit parameters every **182 days**. Keep parameters fixed between refits.
2. **Forecasting.** At origin T, issue $h = 1, \dots, 90$ forecasts:

$$\widehat{\mathbf{f}}_{T+h|T} = \widehat{\mathbf{A}}^{\,h}\widehat{\mathbf{f}}_T, \qquad \widehat{\phi}_{i,T+h|T} = \widehat{\omega}_i + \widehat{\boldsymbol{\beta}}_i'\mathbf{z}_{T+h} + \widehat{\boldsymbol{\Lambda}}_i'\widehat{\mathbf{f}}_{T+h|T}.$$

   $\widehat{\mathbf{f}}_T$: factor state at the origin; $\mathbf{z}_{T+h}$: spline regressors.

   No future score innovations are inserted: under the model, $\mathbb{E}[\tilde{\mathbf{u}}_{T+j} \mid \mathcal{F}_T] = \mathbf{0}$, $j \geq 1$.

3. **Move to the next forecast block.** At $T + 90$, feed the **actual observation** into the filter to refresh the factor state; then issue the next 1–90-step forecasts.

Forecast origins advance every 90 days; parameters refit every 182 days. Throughout: $k = 1$, $\delta = \frac{1}{2}$.

In sample: pooling trades NLL for calibration

Fitted joint-cGB2 ($k = 1$)

Persistence $\widehat{A}$	0.720
Score response $\widehat{B}$	0.064
Loading median	0.98
range	$[0.055, 1.67]$
Information floor $\widehat{\iota}$	30.7
Total log-likelihood	$-1{,}453{,}264$

A single **persistent** common factor; loadings positive and near unity; tail index at its floor ($\eta_i \approx 2.05$) for $28/88$ stations — **heavy tails**.

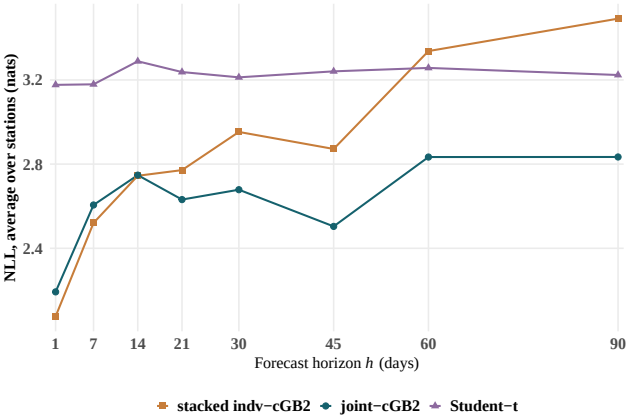
NLL convergence reached.

In-sample scores (average across 88 stations)

	indv-cGB2	joint-cGB2	Student-t
NLL ↓	**2.120**	2.261	3.376
PIT var (ideal $1/12$)	.072	**.075**	.038
Zero-Brier skill ↑	**.215**	.179	.042

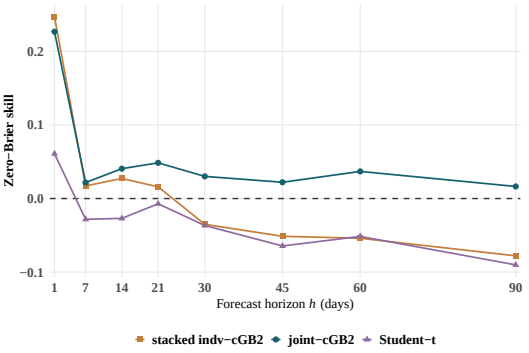
The pooled factor is 0.14 nats per station behind **indv-cGB2** in-sample NLL, while 1.11 nats better than Student-t. The individual fits retain the highest in-sample dry-day skill. **joint-cGB2**is **the best calibrated** among the three models

Out of sample: average NLL turns in favor of pooling after $h = 14$



- At $h = 1$, **indv-cGB2** leads (2.08 vs. 2.19 for **joint-cGB2**).
- **joint-cGB2** moves ahead of **indv-cGB2** from $h = 15$ onward ($h = 15$: 2.69 vs. 2.72; $h = 30$: 2.68 vs. 2.95).
- At $h = 90$, **joint-cGB2** is at 2.83, compared with 3.49 for **indv-cGB2** and 3.22 for Student-t.

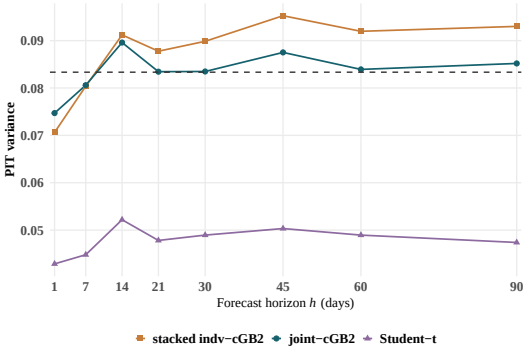
Out of sample: the best zero brier skill, the only positive in long horizons



	h	indv	joint	Stud-t
Zero-Brier skill	1	**.247**	.227	.061
	14	.027	**.041**	$-.027$
	30	$-.035$	**.030**	$-.037$
	90	$-.078$	**.016**	$-.090$

- **joint-cGB2** is the only model with positive Zero-Brier skill at $h = 30, 60, 90$; its skill is positive at 72 of 90 horizons overall.

Out of sample: closest to the PIT-variance target



	h	indv	joint	Stud-t
PIT var (target $1/12$)	1	.071	**.075**	.043
	14	.091	**.090**	.052
	30	.090	**.083**	.049
	90	.093	**.085**	.047

- **joint-cGB2** is closest to the $1/12$ PIT-variance target at 88 of 90 horizons; Student-t variance stays well below target.

Contributions

1. **The first score-driven dynamic factor model with a censored, atomic, non-elliptical measurement** — closed-form filter, a genuine joint law.
2. **Bounded score** $\Rightarrow$ compact factor space $\Rightarrow$ no moment conditions on the data.
3. **Fisher scaling buys identification**: a single scale normalization; orthogonal factors.
4. **Stationarity of a discontinuous DGP** via a mean (Wasserstein) contraction — a technique that could transfer beyond this model.
5. **On UK rainfall:** best-calibrated joint density forecasts, advantage that grows with the horizon, and the only model keeping dry-day skill to a month and above.

Extensions

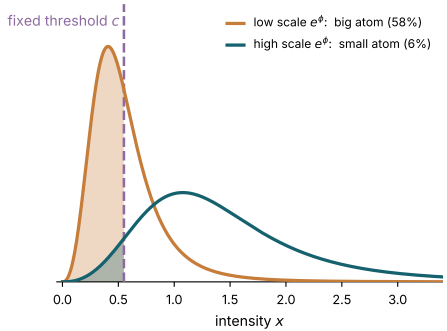
- High-dimensional regime ($N \to \infty$) and its inference.
- Identification-robust inference for weakly-informed thresholds c_i.
- Time-varying shapes.
- Stochastic and endogenous covariates.
- Conditional cross-sectional dependence.

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A4, part 1 — why the Fisher information is time-varying

- **Elliptical (Artemova).** ϕ is pure **location/scale**: a shift moves $\ln x_{it}$ bodily but leaves the **shape** unchanged, so $\mathcal{I}_t^f \equiv \mathcal{I}^f$ is constant in t.

- **Censored (ours; Harvey–Ito).** The threshold c_i is fixed in levels. Moving ϕ **slides the law across** c_i, through $b_{c,it} = \{1 + (e^{\phi_{i,t|t-1}}/c_i)^{\upsilon_i}\}^{-1}$, so the shape — and hence $\mathcal{I}_t^f$ — moves with the state, and is time-varying.



As $c_i \to 0$ censoring vanishes and $\mathcal{I}_t^f \to$ the constant $\upsilon_i^2 \xi_i \varsigma_i / (\xi_i + \varsigma_i + 1)$ (elliptical).

back

A4, part 2 — why we need the information floor

1. **We rescale the score.** The update is $\tilde{\mathbf{u}}_t = (\boldsymbol{\mathcal{I}}_t^f)^{-\delta}\boldsymbol{\nabla}_t$, not the raw $\boldsymbol{\nabla}_t$. With $\delta = \frac{1}{2}$ it sets $\mathrm{Var}_{t-1}(\tilde{\mathbf{u}}_t) = \mathbf{I}_k$, giving $\mathbf{B}$ a **constant, comparable meaning** across t and N.

2. **Our information is time-varying.** Unlike the elliptical case, the fixed threshold c_i makes $\boldsymbol{\mathcal{I}}_t^f$ **move with the state** — it is *low* on near-fully-censored (very dry) days, when the panel says little about the factor.

3. **Scaling can misbehave.** Dividing by $(\boldsymbol{\mathcal{I}}_t^f)^{\delta}$ when the information is near zero would **blow the innovation up** and let the factor jump arbitrarily — the recursion would be unstable.

4. **Hence the floor (A4).** Requiring $\boldsymbol{\mathcal{I}}_t^f \succeq \iota \mathbf{I}_k$ keeps $\tilde{\mathbf{u}}_t$ **bounded**, which is exactly what makes the contraction moduli L_δ, M_δ finite — so invertibility and stationarity go through.

Mild and **checkable**: in the UK data $\widehat{\iota} = 33.2$, far from singular. A larger floor $\iota \Rightarrow$ smaller $M_\delta \Rightarrow$ contraction is *easier*.

A4, part 3 — a bounded scaled score

1. **Why the score is bounded.** In the log-scale ϕ is a *location* parameter of $\ln x_{it}$, and its two branches live in a fixed interval:

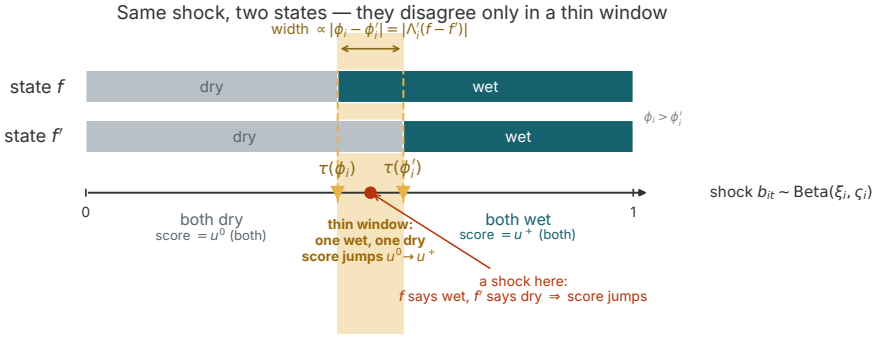
$$-\upsilon_i \xi_i \;<\; u_{it} \;<\; \upsilon_i \varsigma_i, \qquad \mathbb{E}_{t-1}[u_{it}] = 0.$$

   So $\{u_{it}\}$ is a **bounded martingale difference — no tail assumption on** y_{it}.

2. **After scaling.** The scaled innovation is then bounded with the information floor, $\|\tilde{\mathbf{u}}_t\| \le \iota^{-\delta} \|\mathbf{\Lambda}\|_2 \sqrt{N}\, \bar{U}$.

back

Backup: the coupling picture — same shock, two thresholds



- Same shock b_{it} to both states; series i is wet iff b_{it} clears its threshold $\tau(\phi_i)$, which **moves with the state**. Above both $\Rightarrow$ both wet, below both $\Rightarrow$ both dry: same branch, so u_i is smooth.
- They disagree only in the **thin window** between the two thresholds, where u_i jumps $u^0 \to u^+$; its width $\propto |\phi_i - \phi_i'| = |\mathbf{\Lambda}_i'(\mathbf{f} - \mathbf{f}')|$.

back

Backup: observational-equivalence proposition

Statement. $\boldsymbol{\Theta}, \widetilde{\boldsymbol{\Theta}} \in \mathcal{P}$ are observationally equivalent iff the static parameters coincide and $\widetilde{\mathbf{f}}_{t|t-1} = M^{-1}\mathbf{f}_{t|t-1}$ with

$$\delta = 0: \quad \widetilde{\boldsymbol{\Lambda}} = \boldsymbol{\Lambda} M, \quad \widetilde{\mathbf{A}} = M^{-1}\mathbf{A}M, \quad \widetilde{\mathbf{B}} = M^{-1}\mathbf{B}(M')^{-1}, \quad \det M \neq 0;$$

$$\delta > 0: \quad \widetilde{\boldsymbol{\Lambda}} = c\,\boldsymbol{\Lambda} Q, \quad \widetilde{\mathbf{A}} = Q'\mathbf{A}Q, \quad \widetilde{\mathbf{B}} = c^{2\delta-2}Q'\mathbf{B}Q, \quad c > 0,\ Q'Q = \mathbf{I}_k.$$

Admissible M. $\mathrm{GL}(k)$ for $\delta = 0$; the conformal group $\{cQ\}$ for $\delta > 0$. Under distinct factor dynamics $M = PSD$ (resp. cPS); the normalizations (N1)–(N3) remove exactly the permutation P, the signs S and the scale D/c.

back

Backup: the normalizations (N1)–(N3)

After Fisher scaling ($\delta > 0$) and distinct dynamics (A6(ii)), the model is identified **up to a signed permutation and a scale** of the factors — a relabelling $M = c\,P\,S$, where

- P — a **permutation** (which factor is "factor 1"); S — ± 1 **sign flips** (negate a factor and its loadings, no effect); c — an overall **scale** (shrink $\mathbf{f}$, inflate $\mathbf{\Lambda}$; likelihood unchanged).

N1–N3 remove these three, one each:

(N1) **Order** the k factors by convention (decreasing persistence $a_1 > a_2 > \cdots$) — removes P.

(N2) **Sign** each factor by convention (a designated loading made positive) — removes S.

(N3) **Scale** by $\|\mathbf{\Lambda}\|_F = \sqrt{Nk}$ (loadings of unit average squared magnitude) — removes c.

Innocuous labelling conventions — fitted densities, factors and forecasts are unchanged. For $k = 1$ ordering is vacuous: a **sign** and a **scale** suffice.

back

Backup: identification (Proposition; proof in App.)

Shapes, thresholds, intercepts and covariates are pinned down by the conditional law. The dynamic triple $(\mathbf{\Lambda}, \mathbf{A}, \mathbf{B})$ is free only up to a change of basis $\mathbf{f} \mapsto M^{-1}\mathbf{f}$:

Condition	admissible M	cont. d.o.f.
raw score ($\delta = 0$)	$\mathrm{GL}(k)$	k^2
Fisher scaling ($\delta > 0$)	conformal $\{cQ\},\ Q'Q = \mathbf{I}_k,\ c > 0$	$1 + \frac{k(k-1)}{2}$
+ distinct dynamics (A6(ii))	signed permutations $\times$ scale	1
+ scale norm. (N3) $\|\mathbf{\Lambda}\|_F = \sqrt{Nk}$	signed permutations	0
+ order & sign (N1, N2)	identity	0

- Fisher scaling bites because $(\mathcal{I}_t^f)^{-\delta}$ is **equivariant only under** cQ when $\delta > 0$ — the payoff of the time-varying information.
- **No orthogonality restriction** on $\mathbf{\Lambda}$ (unlike the elliptical case); factors uncorrelated by construction. In the application $k = 1$: a scale and a sign suffice.

obs. equivalence back

Backup: cross-sectional dependence & robust inference

A1 makes $\mathcal{L}_T = \sum_t \sum_i \ln g_{it}$ the *exact* panel log-likelihood. If the series are dependent given $\mathcal{F}_{t-1}$:

- $\mathcal{L}_T$ is read as a **composite (pseudo) likelihood** — a sum of the correct marginal conditional densities, omitting only the cross-sectional copula.
- Its score is still a **martingale difference**, zero-mean at $\boldsymbol{\Theta}_0$, so the **QMLE** $\widehat{\boldsymbol{\Theta}}_T$ stays **consistent and asymptotically normal**.
- The information-matrix equality fails $\Rightarrow$ a **sandwich** variance,

$$\sqrt{T}(\widehat{\boldsymbol{\Theta}}_T - \boldsymbol{\Theta}_0) \xrightarrow{d} N\big(\mathbf{0},\ \mathcal{H}^{-1}\mathcal{V}\mathcal{H}^{-1}\big), \qquad \mathcal{V} = \lim_T \tfrac{1}{T}\textstyle\sum_t \mathbb{E}[\mathbf{s}_t\mathbf{s}_t'],$$

  $\mathbf{s}_t$ the time-t score, $\mathcal{V}$ estimated by a HAC / cluster estimator.
- **Filtered factors and forecasts are unchanged** — only the standard errors widen.

back

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